THE SUM OF $f(\gcd(i,n)/\gcd(i,k,n))$ OVER $1 \le i,k \le n$ IS THE DIRICHLET CONVOLUTION OF $f\varphi$ AND A057660 (SCHULTE'S CONJECTURE IN OEIS A350900)

MATH0

Abstract. For an arithmetic function f and $n \ge 1$ let $a(f;n)$ be the sum of $f(\gcd(i,n)/\gcd(i,k,n))$ over $1 \le i,k \le n$. We prove the conjecture recorded by Werner Schulte in OEIS A350900 in 2022: $a(f;n)$ is the Dirichlet convolution of $f\varphi$ with A057660, where $\mathrm{A057660}(m) = \sum_{k=1}^{m} m/\gcd(m,k)$; consequently $a(f;\cdot)$ is multiplicative whenever f is, and the row sums of the triangle A350900 are the Dirichlet convolution of $\mathrm{A002618}(n) = n\varphi(n)$ and A057660. The proof is elementary and rests on Cesàro's classical identity. To our knowledge no proof has been published, and the OEIS still lists the statement as a conjecture, although the row-sum formula had been recorded without proof in A373059; we classify the result as new, but a specialist would regard the argument as routine. The theorem is verified in Lean 4 with Mathlib against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the OEIS data.

1. Introduction

The OEIS entry A350900 [11], contributed by Werner Schulte on January 21, 2022, is the triangle

$$T(n,k) = \sum_{i=1}^{n} \frac{\gcd(i,n)}{\gcd(\gcd(i,k),n)} \qquad (1 \le k \le n),$$

read by rows: $1;\ 3,2;\ 5,5,3;\ 8,5,8,4;\ \ldots$. In the COMMENTS section Schulte replaces the ratio $\gcd(i,n)/\gcd(\gcd(i,k),n)$ by its image under an arbitrary arithmetic function f and sums over both indices, defining

$$a(f;n) = \sum_{i=1}^{n}\sum_{k=1}^{n} f\left(\frac{\gcd(i,n)}{\gcd(\gcd(i,k),n)}\right) \qquad (n \ge 1). \tag{1}$$

He conjectured that $a(f;n)$ is the Dirichlet convolution of $f\varphi$ with A057660 [9], the sequence $\mathrm{A057660}(m) = \sum_{k=1}^{m} m/\gcd(m,k)$; that $a(f;\cdot)$ is multiplicative when f is; and, in the FORMULA section, that the row sums of the triangle are the Dirichlet convolution of $\mathrm{A002618}(n) = n\varphi(n)$ [8] and A057660. The exact text is quoted in Section 2. We prove all of it (Theorem 1).

Method. With $g = \gcd(i,n)$ we have $\gcd(\gcd(i,k),n) = \gcd(g,k)$, which has period $g \mid n$ in k. Grouping k by $\gcd(g,k)$ within one period, and i by $\gcd(i,n)$, uses Cesàro's identity

$$\sum_{k=1}^{n} F(\gcd(k,n)) = \sum_{d \mid n} F(d)\,\varphi(n/d), \tag{2}$$

see formula (22) in Section 2 of Tóth's survey [7] and Lemma 1 of Bordellès [2]. The result is $a(f;n) = \big((f\varphi) * \mathbf{1} * (\mathrm{id}\varphi)\big)(n)$, and the same identity gives $\mathrm{A057660} = \mathbf{1} * (\mathrm{id}\varphi)$.

Date: October 24, 2026.

Status and prior work. In A350900 (revision #13, May 24, 2024, read from a local OEIS mirror of October 17, 2026, and still the current revision on oeis.org on October 25, 2026) both statements are still labeled "Conjecture". The row sums form A373059 [12] (S. Manyama, May 21, 2024), whose FORMULA section states, without proof and without a conjecture label, $\sum_{d|n} \varphi(n/d)(n/d)\,\sigma_2(d^2)/\sigma(d^2)$; since A057660$(d) = \sigma_2(d^2)/\sigma(d^2)$ (a formula of Labos Elemer, 2001, recorded in A057660), this is part (4) of Theorem 1. A. Eldar added there (May 27, 2024) the values at prime powers and the Dirichlet series $\zeta(s)\zeta(s-2)^2/\zeta(s-1)^2$, which also follows from part (4) (Remark 9). Schulte's A350156 [10] makes the analogous statement for $f(n/\gcd(\gcd(i,k),n))$, and Manyama's sequences such as A373060 [13] (May 2024) concern sums of $\gcd(x_1,n)/\gcd(x_1,\dots,x_r,n)$ over $1 \le x_1,\dots,x_r \le n$ (here $r=3$); the OEIS gives formulas for these sequences without proof. Multivariable generalizations of (2) are studied by Bordellès [2] and in a note of P. Bala attached to A368743 [1]. Bala states $\sum_{k_1,\dots,k_r=1}^{n} f(n/\gcd(k_1,\dots,k_r,n)) = \sum_{d|n} f(d)J_r(d)$, with J_r Jordan's totient, and proves it for $r=2$, remarking that this case is Schulte's statement in A350156. The case $r=1$, which he records as the companion of (2), is $\sum_{k=1}^{g} f(g/\gcd(k,g)) = \sum_{e|g} f(e)\varphi(e)$. This is the computation in Step 3 of Lemma 6, and Bala lists A057660 as its case $f = \mathrm{id}$ (Lemma 5). His note does not treat the ratio $\gcd(i,n)/\gcd(i,k,n)$ or A350900. The math0 referees found no proof or counterexample in the OEIS, on arXiv (Fried's paper [4] proving OEIS conjectures does not treat A350900), on SeqFan, the OEIS wiki, or in the gcd-sum literature [7, 2, 1]; the statement is not in formal-conjectures [3] at the commit we use. We therefore classify the result as new, with two caveats: part (4) was already asserted in A373059, and parts (1)–(2) follow from (2) in a few lines, so they may well be known to specialists.

2. Statement

An *arithmetic function* with values in a commutative semiring R is a map $f\colon \mathbb{N} \to R$ with $f(0) = 0$ (Mathlib's convention; the value at 0 never enters (1), since the argument of f there is at least 1). The *Dirichlet convolution* is $(f * g)(n) = \sum_{de=n} f(d)g(e)$, so $(f*g)(0) = 0$; the *pointwise product* is $(fg)(n) = f(n)g(n)$; f is *multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m,n) = 1$. We write $\mathbf{1}$ for the function with $\mathbf{1}(n) = 1$ for $n \ge 1$, $\mathrm{id}(n) = n$, and φ for Euler's totient. We extend (1) by $a(f;0) = 0$, so that $a(f;\cdot)$ is an arithmetic function.

The COMMENTS and FORMULA sections of A350900 [11] contain, verbatim:

> Conjecture: Let f be an arbitrary arithmetic function. Define for n > 0 the sequence a(f; n) = Sum_{i=1..n, k=1..n} f(gcd(i,n)/gcd(gcd(i,k),n)); a(f; n) equals Dirichlet convolution of f(n)*A000010(n) and A057660(n); if f is multiplicative, then a(f; n) is multiplicative; row sums of this triangle use f(n) = n (see formula section).
> Conjecture: Row sums equal Dirichlet convolution of A002618 and A057660.

Neither line carries its own attribution; both date from the entry's creation by Werner Schulte on January 21, 2022 (its author line). Here A000010 is φ.

Theorem 1. *Let* A057660$(m) = \sum_{k=1}^{m} m/\gcd(m,k)$ *and* A002618$(m) = m\varphi(m)$.

(1) *For every arithmetic function* $f\colon \mathbb{N} \to \mathbb{C}$ *and every* $n \ge 1$,

$$a(f;n) = \big((f\varphi) * \mathrm{A057660}\big)(n) = \sum_{d|n} f(d)\varphi(d)\,\mathrm{A057660}(n/d).$$

(2) *If* $f\colon \mathbb{N} \to \mathbb{C}$ *is multiplicative, then so is* $a(f;\cdot)$.
(3) *For every* $n \ge 1$, $\sum_{k=1}^{n} T(n,k) = a(\mathrm{id};n)$.
(4) *For every* $n \ge 1$, $\sum_{k=1}^{n} T(n,k) = (\mathrm{A002618} * \mathrm{A057660})(n)$.

Part (3) is the reading of "row sums of this triangle use f(n) = n"; it is immediate, but it is part of the conjecture's text.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A350900.lean`, reviewed for fidelity, and frozen (SHA-256 `814a4e45...9fb09b`) before any proof attempt. Its definitions and the statement read as follows (verbatim; the copyright comment, the import, the module docstring, docstrings, section headers, blank lines, the sequence `a` with its index map `pos`, the test function `fTest` and all theorems are omitted):

```
namespace Math0.Statements.A350900
open ArithmeticFunction
def T (n k : ℕ) : ℕ :=
  ∑ i ∈ Finset.Icc 1 n, Nat.gcd i n / Nat.gcd (Nat.gcd i k) n
def phi : ArithmeticFunction ℕ := ⟨Nat.totient, Nat.totient_zero⟩
def A057660 (m : ℕ) : ℕ := ∑ k ∈ Finset.Icc 1 m, m / Nat.gcd m k
def A057660Fn : ArithmeticFunction ℕ := ⟨A057660, by simp [A057660]⟩
def A002618 (n : ℕ) : ℕ := n * Nat.totient n
def A002618Fn : ArithmeticFunction ℕ := ⟨A002618, by simp [A002618]⟩
def aF {R : Type*} [AddCommMonoid R] (f : ArithmeticFunction R) (n : ℕ) : R :=
  ∑ i ∈ Finset.Icc 1 n, ∑ k ∈ Finset.Icc 1 n, f (Nat.gcd i n / Nat.gcd (Nat.gcd i k) n)
def aFun {R : Type*} [AddCommMonoid R] (f : ArithmeticFunction R) : ArithmeticFunction R :=
  ⟨aF f, by simp [aF]⟩
def statement : Prop :=
  (∀ f : ArithmeticFunction ℂ, ∀ n : ℕ, 0 < n →
      aF f n = (f.pmul (phi : ArithmeticFunction ℂ) * (A057660Fn : ArithmeticFunction ℂ)) n) ∧
  (∀ f : ArithmeticFunction ℂ, f.IsMultiplicative → (aFun f).IsMultiplicative) ∧
  (∀ n : ℕ, 0 < n →
      ((∑ k ∈ Finset.Icc 1 n, T n k : ℕ) : ℂ) =
        aF ((ArithmeticFunction.id : ArithmeticFunction ℕ) : ArithmeticFunction ℂ) n) ∧
  (∀ n : ℕ, 0 < n → ∑ k ∈ Finset.Icc 1 n, T n k = (A002618Fn * A057660Fn) n)
end Math0.Statements.A350900
```

The four conjuncts are parts (1)–(4) of Theorem 1. `ArithmeticFunctionR` is Mathlib's type of maps $\mathbb{N} \to R$ vanishing at 0, with Dirichlet convolution `*`, pointwise product `pmul` and `IsMultiplicative` as defined above; an ascription such as `(phi : ArithmeticFunction ℂ)` applies the coercion $g \mapsto (n \mapsto (g(n) : \mathbb{C}))$. The division in `T` and `aF` is exact, since $\gcd(\gcd(i,k),n)$ divides $\gcd(i,n)$, and `T`, `A057660` and `A002618` transcribe the NAME lines of the three OEIS entries, not conjectured formulas. Reading "arbitrary arithmetic function" as complex-valued is the most general natural reading; the proof works over any commutative semiring (Lemma 7). The statement file checks the encoding against the OEIS data by evaluation in the kernel (`decide +kernel`): `T_data` reproduces the 12 listed rows of the triangle, `a_data` the 78 listed terms of A350900 read by rows (offset 1), `A057660_data` the 52 listed terms of A057660, and `A002618_data` the first 20 terms of A002618. Two further sanity checks confirm part (4) for $n \le 12$ (`statement_check_rows`) and the identity of part (1) for $n \le 12$ with the non-multiplicative test function $f(m) = 2m + (m \bmod 3)$ computed in $\mathbb{N}$ (`statement_check_aF`).

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A350900.lean`; the corresponding Lean declaration is named with each step. Throughout, M is a commutative additive monoid, R a commutative semiring, and for $c \in \mathbb{N}$, $x \in M$ we write $c \cdot x$ for the c-fold sum. We use $\gcd(0,n) = n$.

Lemma 2 (`sum_Icc_eq_sum_range`)**.** *Let $F\colon \mathbb{N} \to M$ and $n \geq 0$ with $F(0) = F(n)$. Then $\sum_{k=1}^{n} F(k) = \sum_{k=0}^{n-1} F(k)$.*

Proof. Both sides are empty for $n = 0$. For $n = m+1$, the left side is $\sum_{k=0}^{m-1} F(k+1) + F(m+1)$ and the right side is $F(0) + \sum_{k=0}^{m-1} F(k+1)$. □

Lemma 3 (Cesàro; `sum_range_gcd`)**.** *Let $n \geq 1$ and $G\colon \mathbb{N} \to M$. Then*

$$\sum_{i=0}^{n-1} G(\gcd(n,i)) = \sum_{d \mid n} \varphi(n/d) \cdot G(d).$$

Proof. Group the terms by the value of $\gcd(n,i)$, a divisor of n (`Finset.sum_fiberwise_of_maps_to`). For $d \mid n$, the $i \in [0,n)$ with $\gcd(n,i) = d$ are the $i = dj$ with $0 \leq j < n/d$ and $\gcd(n/d, j) = 1$, since $\gcd(n, dj) = d\gcd(n/d, j)$; there are $\varphi(n/d)$ of them (`Nat.totient_div_of_dvd`). □

Lemma 4 (periodicity; `sum_range_mul_gcd`)**.** *Let $g, m \geq 0$ and $F\colon \mathbb{N} \to M$. Then $\sum_{k=0}^{gm-1} F(\gcd(g,k)) = m \cdot \sum_{k=0}^{g-1} F(\gcd(g,k))$.*

Proof. Induction on m; the case $m = 0$ is trivial. Split $[0, g(m+1))$ into $[0, gm)$ and $gm + [0, g)$ and use $\gcd(g, gm+k) = \gcd(g,k)$ (`Nat.gcd_mul_left_add_right`). □

Lemma 5 (`A057660Fn_eq`, `A057660Fn_mult`)**.** *As arithmetic functions,* A057660 $= \mathbf{1} * (\mathrm{id}\varphi)$*, that is,* A057660$(m) = \sum_{d \mid m} d\varphi(d)$ *for $m \geq 1$ (as recorded in* [9]*). In particular* A057660 *is multiplicative.*

Proof. At $m = 0$ both sides vanish. Let $m \geq 1$ and $F(k) = m/\gcd(m,k)$; then $F(0) = m/m = F(m)$, so Lemma 2 and then Lemma 3 with $G(d) = m/d$ give

$$\text{A057660}(m) = \sum_{k=0}^{m-1} \frac{m}{\gcd(m,k)} = \sum_{d \mid m} \varphi(m/d)\, \frac{m}{d} = \sum_{e \mid m} \varphi(e)\, e,$$

where the last step re-indexes by the involution $d \mapsto m/d$ of the divisors of m (`Nat.sum_div_divisors`). The right side is $(\mathbf{1} * (\mathrm{id}\varphi))(m)$ (`zeta_mul_apply`). Since $\mathbf{1}$, id and φ are multiplicative and pointwise products and Dirichlet convolutions of multiplicative functions are multiplicative, so is A057660. □

Lemma 6 (`aF_eq`)**.** *Let $f\colon \mathbb{N} \to R$ and $n \geq 1$. Then*

$$a(f;n) = \sum_{g \mid n} \varphi(n/g) \cdot \Big(\frac{n}{g} \cdot \sum_{e \mid g} \varphi(e) \cdot f(e) \Big).$$

Proof. Step 1: both indices run over $[0,n)$. For fixed i, the summand of (1) has the same value at $k = 0$ and $k = n$, as $\gcd(\gcd(i,0), n) = \gcd(i,n) = \gcd(\gcd(i,n), n)$; the inner sum has the same value at $i = 0$ and $i = n$, as $\gcd(0,n) = n = \gcd(n,n)$ and $\gcd(\gcd(0,k), n) = \gcd(k,n) = \gcd(\gcd(n,k), n)$. Applying Lemma 2 to both sums, and rewriting $\gcd(\gcd(i,k), n) = \gcd(\gcd(n,i), k)$ (`Nat.gcd_right_comm`), we obtain

$$a(f;n) = \sum_{i=0}^{n-1} \sum_{k=0}^{n-1} f\left(\frac{\gcd(n,i)}{\gcd(\gcd(n,i),k)} \right).$$

Step 2: grouping i. The inner sum depends on i only through $g = \gcd(n,i)$, so Lemma 3 with $G(g) = \sum_{k=0}^{n-1} f(g/\gcd(g,k))$ gives

$$a(f;n) = \sum_{g \mid n} \varphi(n/g) \cdot \sum_{k=0}^{n-1} f\left(\frac{g}{\gcd(g,k)} \right).$$

Step 3: the sum over k. Fix $g \mid n$; then $g \geq 1$ and $n = g \cdot (n/g)$. Lemma 4 with $m = n/g$ and $F(d) = f(g/d)$, then Lemma 3 for g with $G(d) = f(g/d)$, then the involution $d \mapsto g/d$ give

$$\sum_{k=0}^{n-1} f\Big(\frac{g}{\gcd(g,k)}\Big) = \frac{n}{g} \cdot \sum_{k=0}^{g-1} f\Big(\frac{g}{\gcd(g,k)}\Big) = \frac{n}{g} \cdot \sum_{d \mid g} \varphi(g/d) \cdot f(g/d) = \frac{n}{g} \cdot \sum_{e \mid g} \varphi(e) \cdot f(e). \quad \square$$

Lemma 7 (`main`)**.** *Let $f\colon \mathbb{N} \to R$ be an arithmetic function and $n \geq 1$. Then $a(f;n) = \big((f\varphi) * \mathrm{A057660}\big)(n)$, where φ and A057660 are mapped into R by the canonical map $\mathbb{N} \to R$.*

Proof. The canonical map from $\mathbb{N}$-valued to R-valued arithmetic functions respects Dirichlet convolution (`natCoe_mul`), so Lemma 5 gives $\mathrm{A057660} = \mathbf{1} * (\mathrm{id}\varphi)$ over R. By associativity of Dirichlet convolution,

$$(f\varphi) * \mathrm{A057660} = \big((f\varphi) * \mathbf{1}\big) * (\mathrm{id}\varphi),$$

and, since $((f\varphi) * \mathbf{1})(g) = \sum_{e \mid g} f(e)\varphi(e)$ (`coe_mul_zeta_apply`) and $(P * Q)(n) = \sum_{g \mid n} P(g)Q(n/g)$ (`Nat.sum_divisorsAntidiagonal`),

$$\big((f\varphi) * \mathrm{A057660}\big)(n) = \sum_{g \mid n} \Big(\sum_{e \mid g} f(e)\varphi(e)\Big) \frac{n}{g}\, \varphi\Big(\frac{n}{g}\Big).$$

By commutativity of R this equals the right side of Lemma 6. $\square$

Lemma 8 (`rowsum_eq`)**.** *For every $n \geq 0$, $\sum_{k=1}^{n} T(n,k) = a(\mathrm{id}; n)$, computed in $\mathbb{N}$.*

Proof. Exchange the two finite sums in $\sum_{k=1}^{n} T(n,k)$ (`Finset.sum_comm`) and compare with (1). $\square$

Proof of Theorem 1 (`result`). (1) is Lemma 7 with $R = \mathbb{C}$.

(2) The arithmetic functions $a(f;\cdot)$ and $(f\varphi) * \mathrm{A057660}$ over $\mathbb{C}$ agree at 0, where both vanish, and at every $n \geq 1$ by part (1). If f is multiplicative, so is $f\varphi$, as φ (and its image in $\mathbb{C}$) is; A057660 and its image in $\mathbb{C}$ are multiplicative by Lemma 5; and Dirichlet convolution preserves multiplicativity (`IsMultiplicative.pmul`, `IsMultiplicative.mul`, `IsMultiplicative.natCast`).

(3) By Lemma 8, and because the map $\mathbb{N} \to \mathbb{C}$ commutes with finite sums and sends $\mathrm{id}(m)$ to m.

(4) By Lemma 8 and Lemma 7 with $R = \mathbb{N}$ and $f = \mathrm{id}$, $\sum_{k=1}^{n} T(n,k) = \big((\mathrm{id}\varphi) * \mathrm{A057660}\big)(n)$; for $R = \mathbb{N}$ the canonical map is the identity (`natCoe_nat`), and $\mathrm{id}\varphi = \mathrm{A002618}$. $\square$

Remark 9. For $\mathrm{Re}\, s > 3$ the Dirichlet series of $\mathrm{A002618} = \mathrm{id}\varphi$ is $\zeta(s-2)/\zeta(s-1)$ and, by Lemma 5, that of A057660 is $\zeta(s)\zeta(s-2)/\zeta(s-1)$. Part (4) therefore gives the Dirichlet series $\zeta(s)\zeta(s-2)^2/\zeta(s-1)^2$ for the row sums, as recorded by Eldar in A373059 [12]. This consequence is not formalized.

4. Machine verification

Files and declarations. The proof is in `lean/Math0/Frontier/OEIS/A350900.lean` in the math0 repository, `https://github.com/Foundation-0/math0`; it imports the frozen statement file and `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A350900.result`, of type `Math0.Statements.A350900.statement`. The private lemmas are named in Section 3, plus `sum_Icc_eq_sum_range_succ`, `phi_apply` and `phi_mult`. The statement of `main` and the end of the file are, verbatim except that the proof of `main`, docstrings, blank lines and the last 14 lines of the proof of `result` (marked `-- ...`; the three remaining parts, as in Section 3) are omitted:

```
private lemma main {R : Type*} [CommSemiring R] (f : ArithmeticFunction R) (n : ℕ)
    (hn : 0 < n) :
    aF f n = (f.pmul (phi : ArithmeticFunction R) * (A057660Fn : ArithmeticFunction R)) n
theorem result : statement := by
  refine ⟨fun f n hn => main f n hn, fun f hf => ?_, fun n _ => ?_, fun n hn => ?_⟩ -- ...
end Math0.Frontier.A350900
#math0_settles Math0.Frontier.A350900.result Math0.Statements.A350900.statement
```

The proof file uses no `decide`, `native_decide` or `set_option`; it involves no finite computation and does not use the data checks of the statement file.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (i) `statement` is a definition of type `Prop` without universe parameters; (ii) `statement` and `result` depend on no axioms other than `propext`, `Classical.choice` and `Quot.sound` (excluding `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (iii) the type of `result` is syntactically `statement` or its negation, up to metadata, with no unfolding. It reports

```
math0 audit: 'Math0.Statements.A350900.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A350900.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A350900.result' proves 'Math0.Statements.A350900.statement'
```

Toolchain and reproduction. We use Lean 4 [6] `v4.33.1` and Mathlib [5] `v4.33.1`; the repository also pins formal-conjectures [3] at commit `c77fdd4`, not used here. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib cache, builds all math0 results with `lake build` (re-running every audit), checks that the pinned dependencies are unmodified, and reports the number of passed statement checks. The single file is checked by `lake env lean Math0/Frontier/OEIS/A350900.lean` in `lean/` (under a minute on our machine, once the statement file is built).

References

[1] P. Bala, *GCD sum theorems. Two multivariable Cesàro type identities*, note attached to OEIS A368743, January 2024, `https://oeis.org/A368743/a368743.pdf`.

[2] O. Bordellès, *A multidimensional Cesàro type identity and applications*, J. Integer Seq. **18** (2015), Article 15.3.7.

[3] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An open and evolving benchmark for verified discovery in mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[4] S. Fried, *Further proofs of conjectures from the OEIS*, arXiv:2607.24832, 2026.

[5] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[6] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

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[8] The OEIS Foundation Inc., `https://oeis.org/A002618` ($n\varphi(n)$), entry by N. J. A. Sloane.

[9] The OEIS Foundation Inc., `https://oeis.org/A057660`, entry by H. Gould (2000).

[10] The OEIS Foundation Inc., `https://oeis.org/A350156`, entry by W. Schulte (January 19, 2022).

[11] The OEIS Foundation Inc., `https://oeis.org/A350900`, entry and conjecture by W. Schulte (January 21, 2022); revision #13, May 24, 2024.

[12] The OEIS Foundation Inc., `https://oeis.org/A373059` (row sums of A350900), entry by S. Manyama (May 21, 2024), with formulas by A. Eldar (May 27, 2024); revision #19.

[13] The OEIS Foundation Inc., `https://oeis.org/A373060`, entry by S. Manyama (May 21, 2024).